Arduino && WebSockets

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Check-In

Which Homer are you?

Goal

Use an Arduino to communicate instantly with a Node server using web sockets.

Agenda

- Background 10 min
- Setup 5 min
- Arduino to Arduino 20 min
- Browser to Arduino 20 min
- Questions 5 min

Why Websockets?

Instant communication!

No polling delays

Today

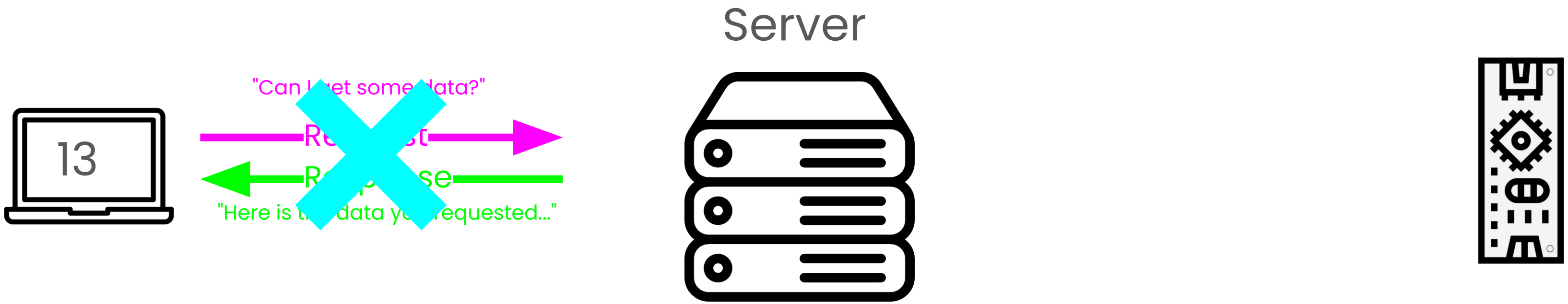
Two simple examples!

1. Arduinos instantly talking to each other through web server, no frontend.
2. Web frontend instantly controlling distributed Arduinos.

How Socket Work

HTTP

What we did last week



data: {[
  "Sensor":13
]}

HTTP

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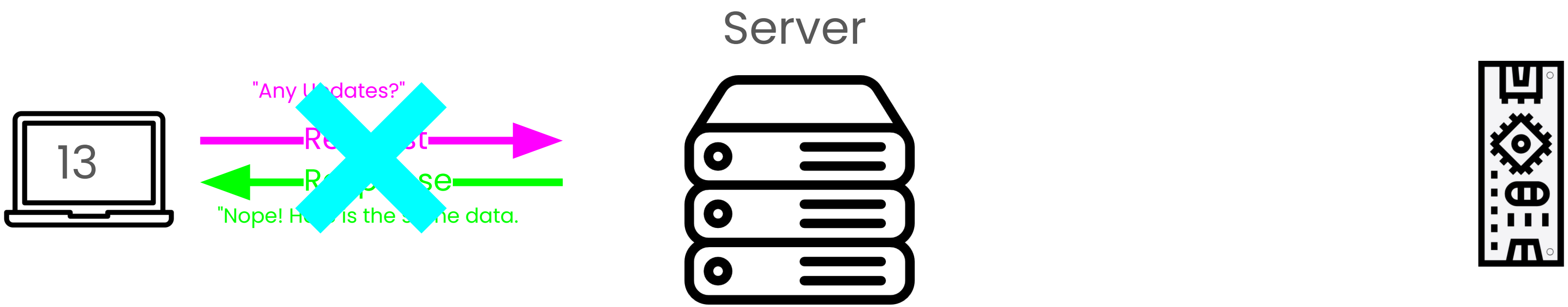

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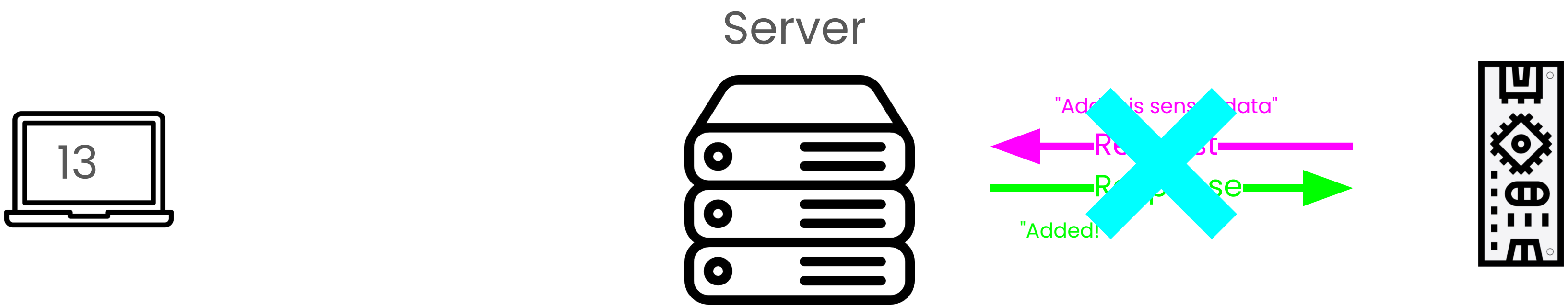


```
data: {[
  "Sensor":13
  "Sensor":505
]}
```

HTTP

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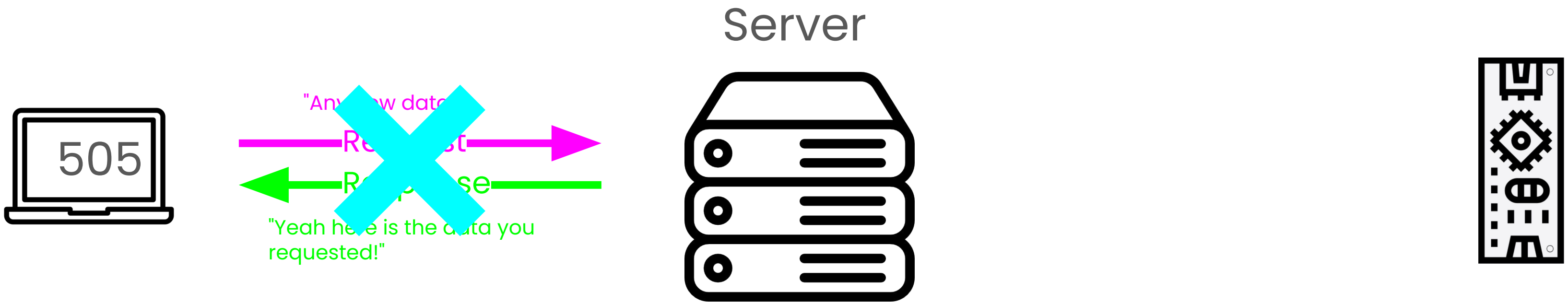


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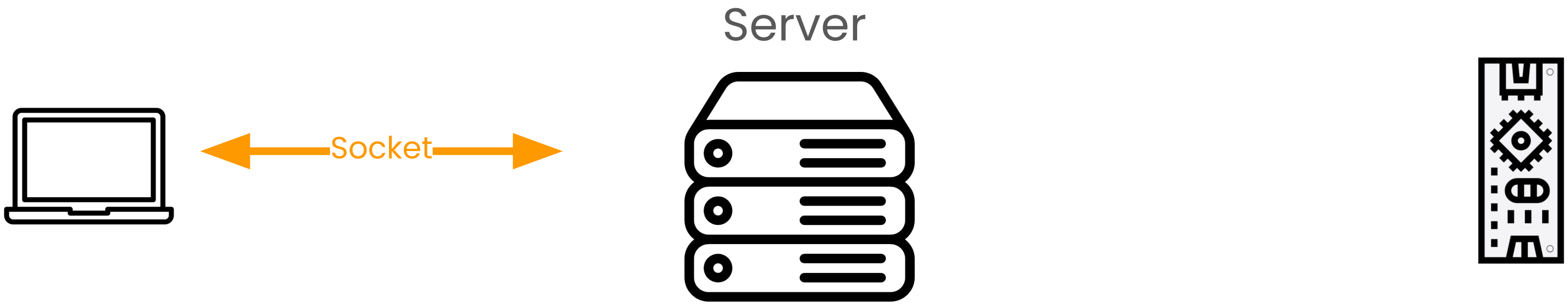
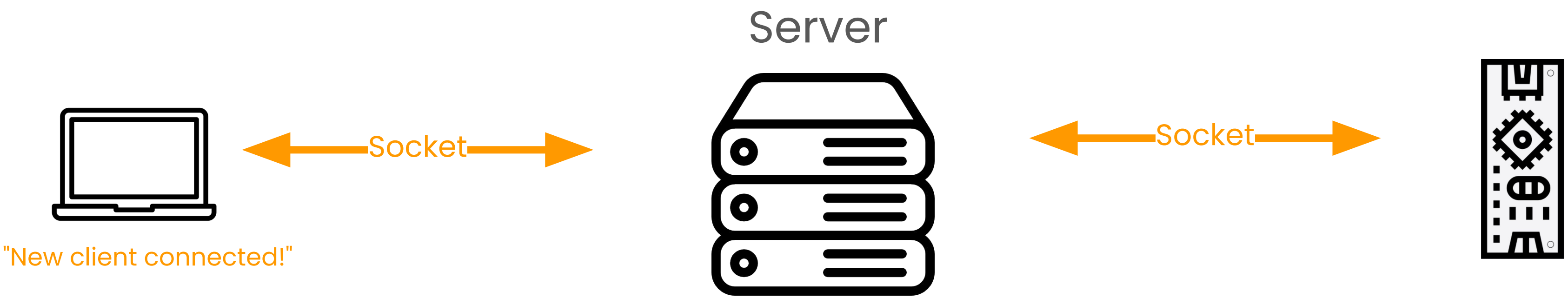
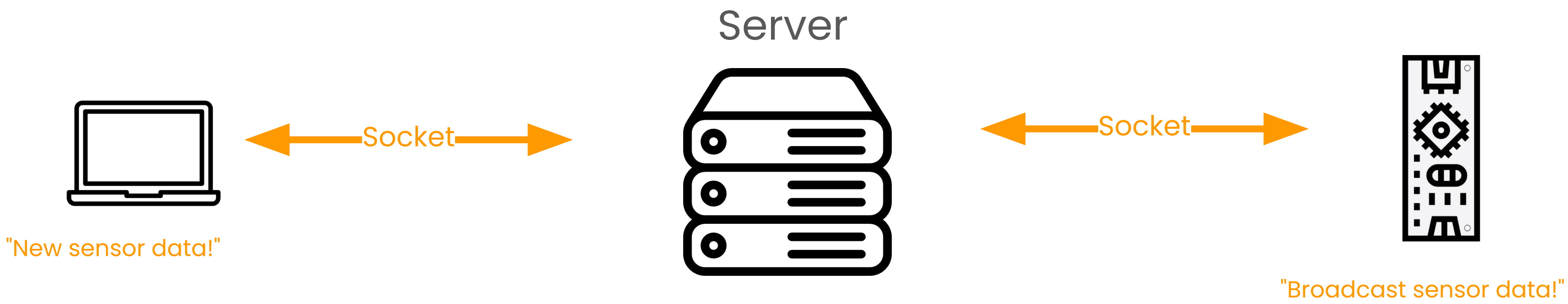
Socket



Socket

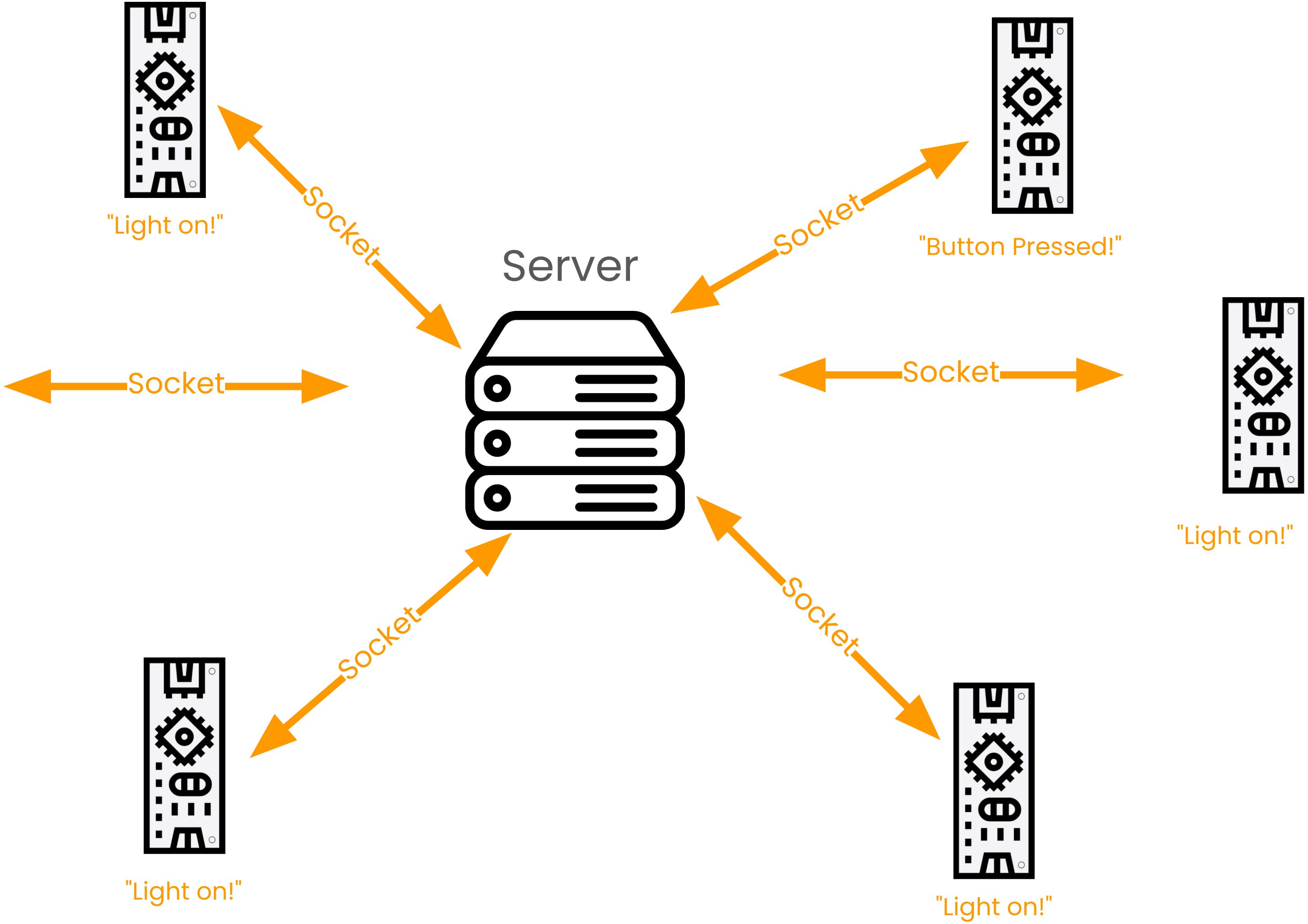


Socket



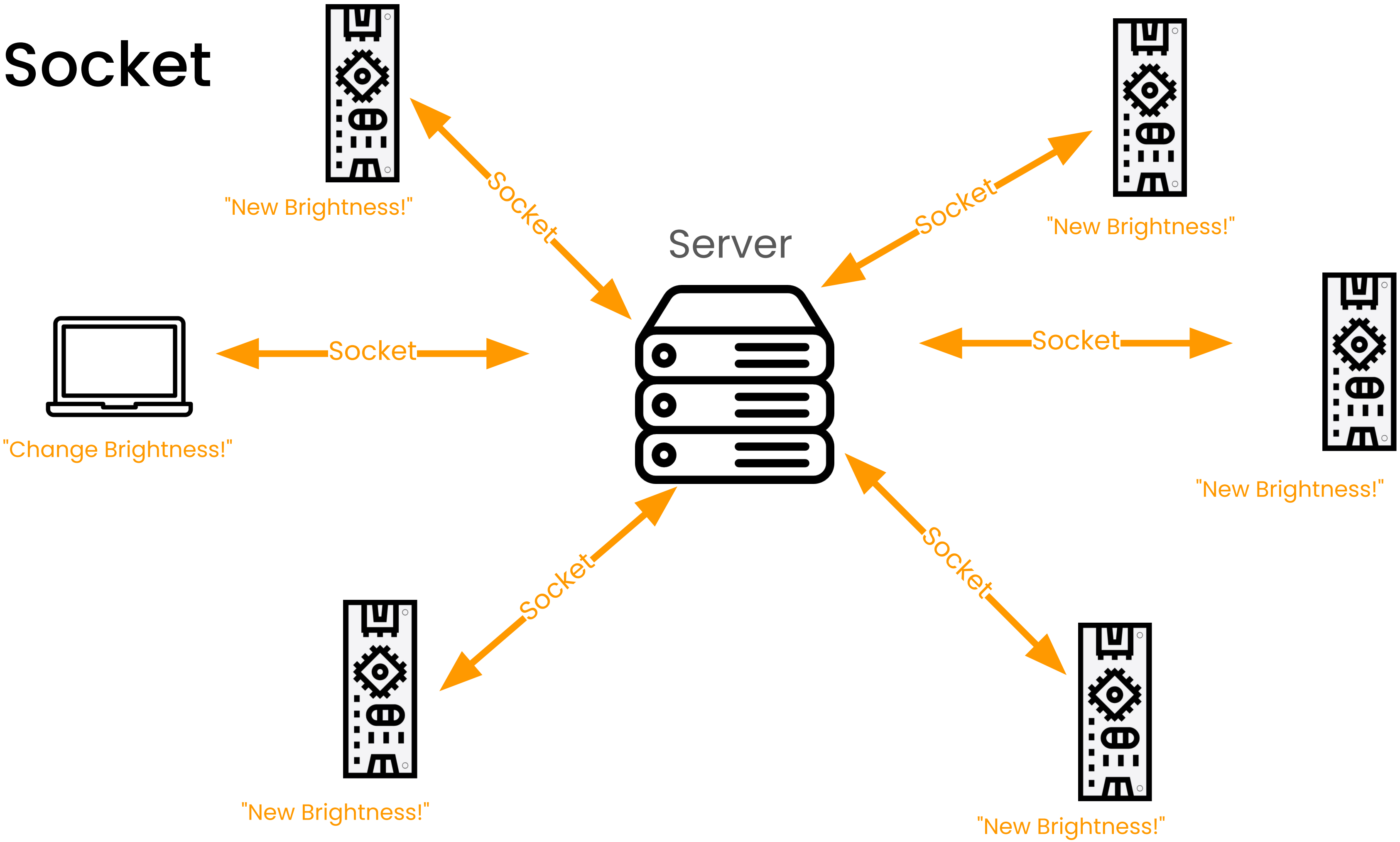
Server
Socket
Socket
"New sensor data!"
"Broadcast sensor data!"

Socket



"Light on!"
Socket
"Button Pressed!"
Server
"Light on!"
Socket
Socket
"Light on!"
Socket
Socket
Socket
"Light on!"
"Light on!"

Socket


"New Brightness!"
Socket
Server
Socket
"New Brightness!"
Socket
Socket
"Change Brightness!"
"New Brightness!"
Socket
Socket
"New Brightness!"
"New Brightness!"

So we need...

- Server with routes + sockets
- Arduino connected to the internet
- A way for the Arduino connect to server socket
- Server handles sockets messages
- Arduino reacts to new messages

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Server

Read through `server.js` file. What do you see?

5 min

Why no socket.io?

Why no socket.io?

socket.io is Node + frontend library for socket library.

It provides some sugar for making sockets easier and has cool features (rooms, polling fallback, etc)

BUT it doesn't work with Arduino without some extra work that isn't worth it.

Instead we will use a plain ol' web socket connection with a different Node library.

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WifiNINA Library

Connect Arduino to Wifi

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WebSocketClient

Arduino Library for Sockets

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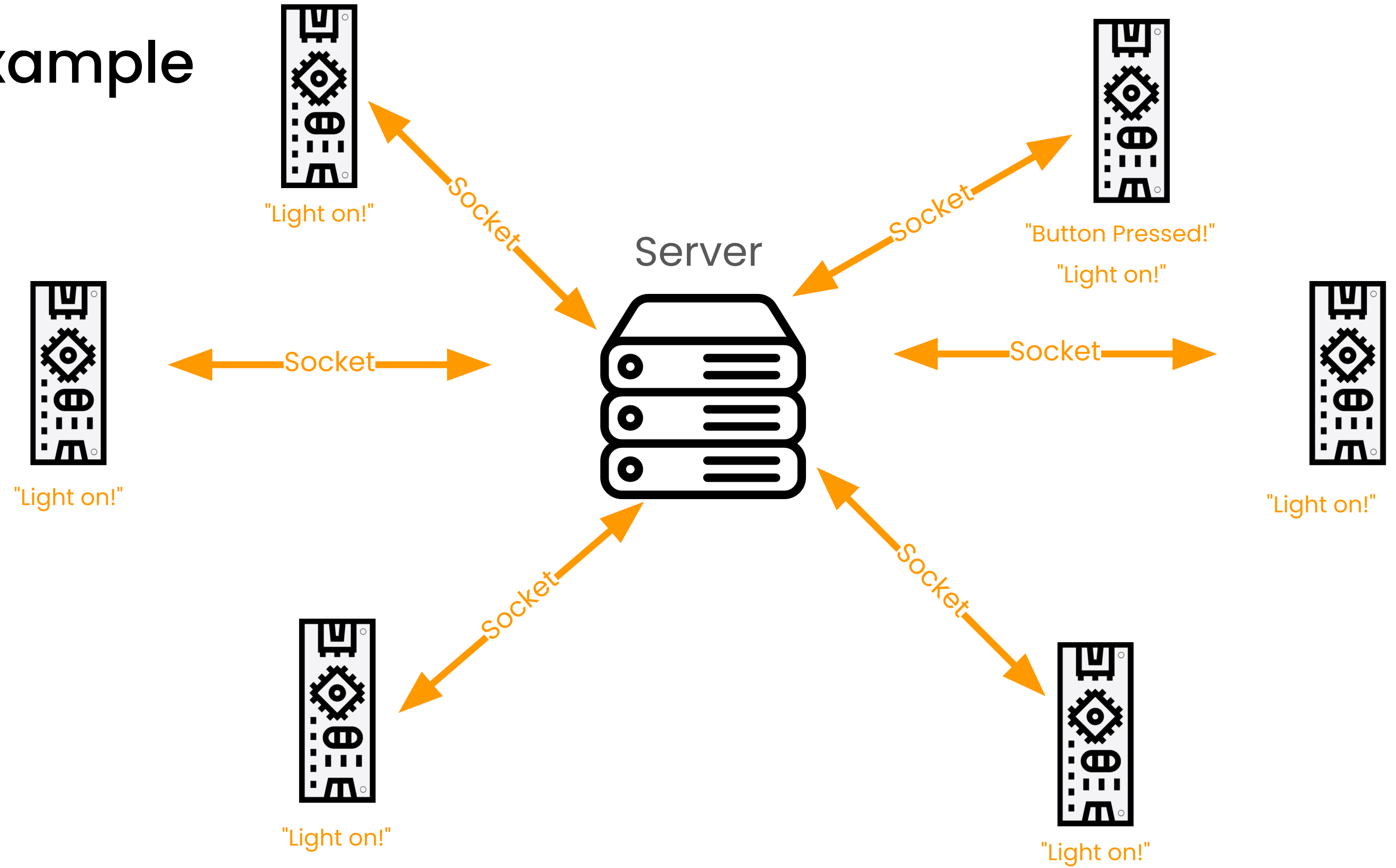
Example 1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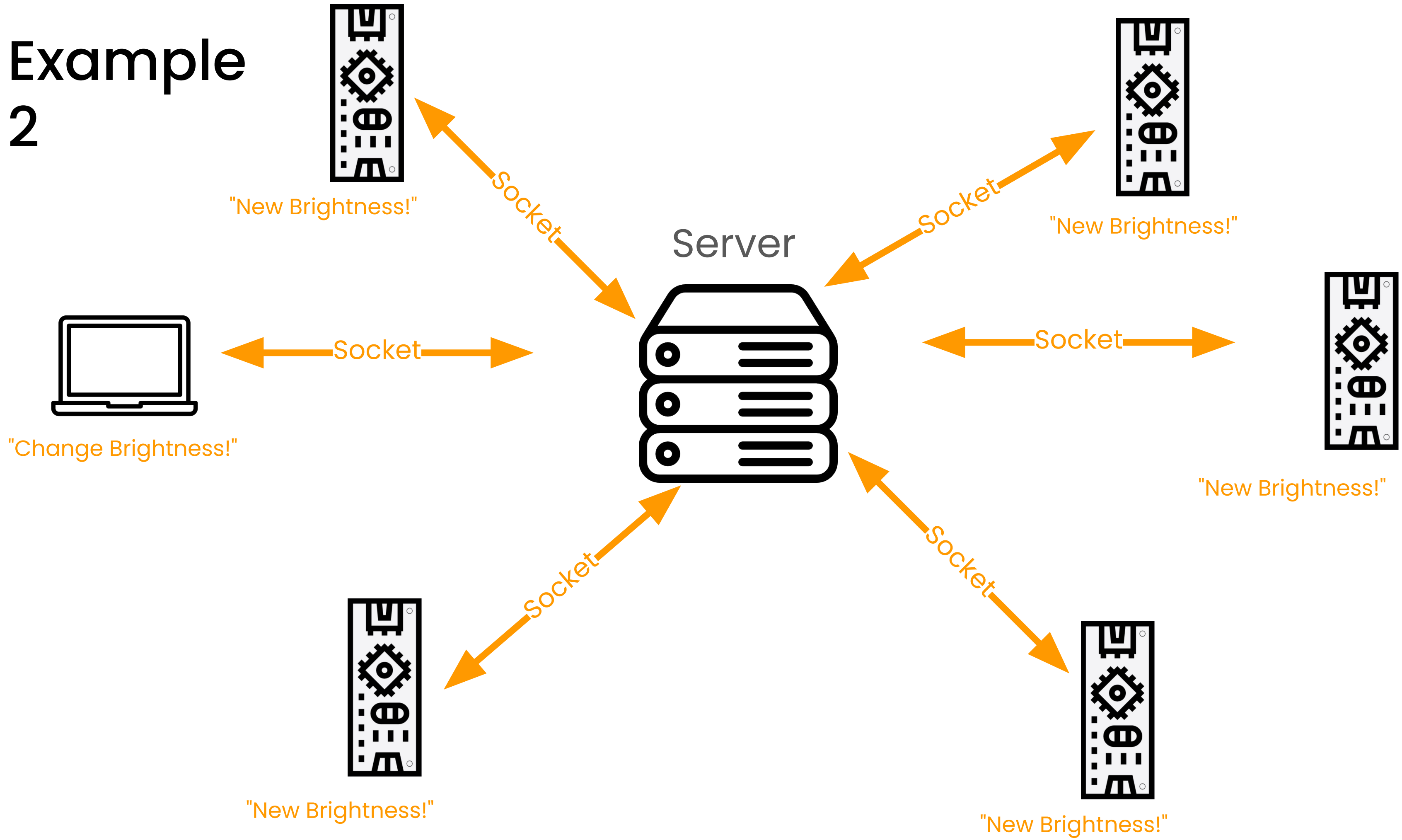
Example 2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Challenge

Combine the two sketches!

Connect a force sensor to the Arduino

Write a p5 sketch the uses the force the change the size of the circle.

The higher the force the smaller the circle.

Hint - You'll either have to change and deploy your own server or highjack one of my value, like brightness

More Ways to Connect

Arduino Connection Options

Method	Range	Power	Built in?
WiFi / HTTP	Short to medium	High	✅
BLE	Short	Very low	✅
LoRa	Long	Low	🚫
ESP-NOW	Medium	Low	🚫 esp32 only
Cellular	Short to very long	Very high	🚫

Questions?